Statistics & Probability

Section 1: Central Tendency & Weighted Averages

1 A student scored $82, 88, 91,$ and 85 on four exams. What score must the student earn on the fifth exam to achieve an overall mean score of 88?

a. 90

b. 92

c. 94

d. 96

2 In a set of 7 integers: $\{4, 7, 9, x, 14, 18, 20\}$ arranged in ascending order, the median is equal to the mean. What is the value of x?

a. 12

b. 13

c. 14

d. 15

3 In a chemistry course, homework counts for 20%, labs count for 30%, and tests count for 50%. If a student earns 90 on homework, 80 on labs, and 86 on tests, what is the weighted final grade?

a. 84

b. 85

c. 86

d. 87

Section 2: Data Spread, Range & Standard Deviation

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4 A data set consists of the numbers $\{3, 7, 8, 12, 15, 18, 21\}$. What is the interquartile range (IQR) of this data set?

a. 9

b. 11

c. 12

d. 14

5 If every value in a data set is multiplied by 3 and then increased by 5, how is the standard deviation σ affected?

a. It increases by 5

b. It is multiplied by 3

c. It is multiplied by 3 and increased by 5

d. It remains unchanged

6 Which of the following data sets has a standard deviation equal to 0?

a. $\{5, 5, 5, 5, 5\}$

b. $\{1, 2, 3, 4, 5\}$

c. $\{-2, 0, 2\}$

d. $\{10, 20, 30\}$

Section 3: Probability & Compound Events

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7 A bag contains 6 red marbles, 8 blue marbles, and 10 green marbles. If one marble is selected at random, what is the probability that it is NOT blue?

a. $\frac{1}{3}$

b. $\frac{2}{3}$

c. $\frac{1}{4}$

d. $\frac{3}{4}$

8 Two fair standard six-sided dice are rolled simultaneously. What is the probability that the sum of the two numbers rolled is greater than 9?

a. $\frac{1}{6}$

b. $\frac{5}{36}$

c. $\frac{1}{9}$

d. $\frac{7}{36}$

9 Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.5$, what is the probability that at least one of the events occurs, $P(A \cup B)$?

a. 0.70

b. 0.80

c. 0.90

d. 0.20

Section 4: Combinations & Permutations

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10 In how many distinct ways can a committee of 4 members be chosen from a group of 9 people?

a. 36

b. 84

c. 126

d. 3024

11 How many different 4-digit identification codes can be formed using digits 0 through 9 if repetition of digits is not allowed and the first digit cannot be 0?

a. $4,536$

b. $5,040$

c. $9,000$

d. $3,024$

12 A restaurant offers 4 appetizers, 6 main courses, and 3 desserts. How many distinct 3-course dinner combinations are possible?

a. 13

b. 36

c. 72

d. 144

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